

Northern Africa and Western Asia Research and Action Agenda

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Executive summary

Climate change is increasingly recognised as a threat to mental health, compounding risks for poor mental health outcomes and destabilising the conditions needed for good mental health. While research at the intersection of climate change and mental health has increased in recent years, the field remains disconnected, uneven and siloed, slowing urgent progress to address the mental health impacts of climate change.

Connecting Climate Minds (CCM) is a Wellcome-funded initiative to cultivate a collaborative, transdisciplinary climate change and mental health field with a clear and aligned vision. Over the last year, we have convened experts across disciplines, sectors, and countries to develop regional and global research and action agendas. These agendas set out 1) research priorities to understand and address the needs of people experiencing the mental health burden of the climate crisis, and 2) priorities to enable this research and translate evidence into action in policy and practice.

This report presents the research and agenda for climate change and mental health in North Africa and Western Asia (NAWA) – a region also often referred to as the Middle East and Northern Africa region (MENA) and referred to throughout this document as NAWA/MENA.

The main challenges to mental health in the context of climate change in NAWA/MENA stem from the direct and indirect impacts of climate-related events. Climate change exacerbates the complex nature of urban areas by influencing factors that drive rural-to-urban migration, leading to denser populations in cities where resources may be strained. This migration is often driven by climate-induced hardships in rural areas, such as extreme temperatures, drought and decreased agricultural productivity. Once in urban settings, individuals face added pressures from the fast-paced urban environment, including increased stress and anxiety. Additionally, climate change-related events like flooding, wildfires, extreme temperatures, and drought can lead to displacement, exacerbate existing mental health issues and create socio-cultural dimensions that impact mental wellbeing. These impacts include heightened stress, depression and changes in lifestyle and social interactions, further challenging the mental health of these individuals.

The CCM project identified several priority research themes for NAWA/MENA covering the following areas:

- **Exploring impacts of climate change on mental health and pathways by which these impacts occur:** Research should analyse the nature and severity of the impacts of climate change on mental health outcomes. It is important to explore the biopsychosocial pathways linking climate change to mental health challenges, including those most prevalent in NAWA/MENA.
- **Identifying the most affected groups:** Research should identify the groups that are most vulnerable to the mental health impacts of climate change. For NAWA/MENA, this includes studying the effects on children, people with disabilities, women and other vulnerable populations.
- **Understanding NAWA/MENA-specific impacts:** Research should explore the specific regional impacts of climate change on mental health. This includes studying the unique challenges faced by different countries, geographies (e.g., coastal or urban communities) and populations and identifying region-specific interventions to address these challenges that account for local cultural needs.
- **Exploring synergies between climate change and mental health policies, particularly for**

vulnerable populations: Research should focus on the integration of mental health considerations into climate change policies and strategies and vice versa. This includes evaluating the effectiveness of policy interventions in protecting mental health and promoting resilience in the face of climate change.

- **Understanding the mental health benefits of climate actions and interventions:** Research should explore the co-benefits of climate adaptation and mitigation actions for mental health (e.g., investing in renewable energy sources and green infrastructure projects).
- **Assessing the accessibility of mental health services:** Research should examine the accessibility of mental health services during climate-related crises. This includes evaluating the availability of psychological first-aid training for healthcare workers and assessing the preparedness of healthcare facilities to address mental health needs during climate-related events.

Realising demonstrable progress requires dedicated leadership, resources and political will alongside sustained, inclusive participation from all sectors of society. This document seeks to inspire by demonstrating the possibilities if we collectively leverage knowledge, technology, policy and community strengths to address the mental health impacts of climate change. The path forward entails surmounting complex obstacles and system transformations requiring persistent effort. By taking ambitious first steps grounded in compassionate optimism, we can move towards a future where the health of both people and the planet is valued and viewed as interconnected.

Introduction

Context

Climate change and mental health are two of our greatest global challenges and awareness of the intersection between mental health and the climate crisis has grown rapidly in recent years.¹ Climate change exacerbates mental health challenges by increasing exposure to extreme heat and the traumas of extreme weather events,² destabilising the conditions needed for good mental health and wellbeing (for example, water and food insecurity, forced migration, polluted air, loss of treasured environments),² disrupting access to healthcare,³ and increasing psychological distress through awareness of climate threats and insufficient climate action.⁴ People living with mental health challenges are also particularly vulnerable to the stressors of the climate crisis, such as increased risk of physical heat stress and death during heatwaves.^{5, 6, 7}

In response to the mounting mental health toll of the climate crisis, research in the climate and mental health field has grown rapidly. Nevertheless, key evidence gaps exist in many regions, including the mental health burden attributable to climate change, the pathways and mechanisms underlying these impacts, the co-benefits of climate action for mental health and the best interventions or solutions to support mental health in a changing climate. Climate change and mental health research remain frustratingly disconnected across disciplines, sectors, and geographies, and are unevenly focused on certain topics and global regions.⁸ Moreover, siloed decision-making slows the translation of evidence to aligned action across climate and mental health policy and practice.^{9, 10} A more inclusive, connected agenda is urgently needed to generate the evidence to truly understand, monitor and respond to the interconnections between climate change and mental health.

Connecting Climate Minds

Connecting Climate Minds (CCM) is a Wellcome-funded project launched in 2023 to develop an inclusive agenda for research and action in climate change and mental health. The project has two key, intertwined aims. The first is to develop an aligned and inclusive agenda for research and action that is grounded in the needs of those with lived experience of mental health challenges in the context of climate change, to guide the field over the coming years. The second is to kickstart the development of connected communities of practice for climate change and mental health in seven global regions (designated by the Sustainable Development Goals), equipped to enact this agenda. We aim to combine the strengths of a global perspective and regional focus and bring together diverse disciplinary perspectives into a shared vision that can ensure research is effective at addressing priority evidence gaps and informing changes in policy and practice at the intersection of climate change and mental health.

Through bringing together experts across diverse disciplines, sectors, and countries, the CCM team has facilitated the development of a lived experience-informed research and action agenda for the climate change and mental health field in North Africa and Western Asia/Middle East and Northern Africa (NAWA/MENA).

Objectives of the research and action agenda

The research and action agenda is designed to focus future efforts to help those who are experiencing or will experience, the compounding mental health challenges of climate change. It aims to support those who are already responding to these challenges – through communities, research, policy, and practice – by building a more connected and collaborative climate change and mental health field. It also aims to empower experts across disciplines and sectors to join and make progress in this area by identifying clear priorities and fostering a more inclusive and transdisciplinary field.

The agenda addresses these aims through four core objectives, which are to:

1. Identify priorities for research that can inform action to meet the needs of people experiencing and responding to the mental health impacts of climate change in NAWA/MENA.
2. Identify what is needed to implement research priorities and translate evidence to action in policy and practice in NAWA/MENA.
3. Build understanding among researchers, funders, and policy experts across disciplines and sectors of their role in furthering climate change and mental health research and equip them with clear and actionable priorities to contribute to this space.

The regional agenda will be integrated with six other regional agendas to inform a **global research and action agenda** for climate change and mental health. This will ensure that global research efforts and investment in climate change and mental health are grounded in regional-level priorities. Importantly, the global agenda will also integrate insights from agendas developed with and for some of the most affected groups globally, namely Indigenous Peoples, youth, small farmers, and fisher peoples.

Use of the terms climate change and mental health

Climate change, mental health and their intersection are complex and wide-ranging fields. For this agenda, we define the scope of these terms as follows.

By **mental health challenges**, we mean thoughts, feelings and behaviours that affect a person's ability to function in one or more areas of life and often involve significant levels of psychological distress. This includes, but is not limited to, anxiety, depression, post-traumatic stress, psychosis, suicidal thoughts and substance abuse.

By **experiences of the effects of climate change**, we mean: 1) experiencing direct impact of climate hazards, such as more frequent and intense heatwaves, wildfires, bushfires, drought, floods or storms (e.g., typhoons, hurricanes, cyclones), and 2) experiencing disruption to the social and environmental determinants of good mental health, such as being forced to move home, not being able to access food or water, losing livelihood or homelands, or disruption to cultural practices because of climate change.

Mental health challenges in the context of climate change include:

- How climate change may lead to worsening pre-existing mental health problems.
- How climate change may contribute to the prevalence or impact of existing mental health problems.
- How climate change may impact treatment access or effectiveness for those with problems.
- How climate change may lead to new mental health problems.

The convening work of CCM presents a key opportunity to build our understanding of diverse perspectives, framings and terminologies in NAWA/MENA, which we have sought to reflect within the research and action agenda.

Background to Connecting Climate Minds

Regional Community of Practice

In NAWA/MENA, CCM is led by a Regional Community Team (RCT), responsible for convening diverse expertise across the region and building regional capacity to create and enact the research and action agenda. The structure of the RCT is outlined below.

Regional Community Convenor (RCC)

Purpose: Responsible for developing and delivering project activities in the region, including convening and supporting a regional community of diverse expertise.

Members: *Jordan Health Aid Society International, Jordan*

Dr. Yaroup Ajlouni

Dr. Duha Al Omari

Dr. Tala Dabbas

Mo'nes Al-Qudah

Co-convenors

Purpose: Bringing additional breadth of expertise across disciplines and countries (i.e. organisations spanning climate expertise, stress neuroscience and mental health expertise, in different sectors), providing technical advice and review, and supporting project delivery.

Members: *Dr. Rouba Katrina, psychiatrist, Syria*

Dr. Nabaz Mirani, Doctors Aid for Medical Activities (DAMA) Org, Iraq

Mazen Malkawi, Regional Advisor on Environmental Health, at the World Health Organization (WHO), Eastern Mediterranean Region Office (EMRO), Regional Center for Environmental Health Action (CEHA), Jordan

Lived experience advisory group (LEAG)

Purpose: Advisory board of experts with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. Drawing on their unique expertise and wisdom, LEAGs provide vital community-centred perspectives and guidance that inform the overarching approach and outputs of the project.

Members: *Ashraf F. ALqudah, Clinical Psychology Associate Professor, Jordan*

Dr Hamad Al Ghafri, Consultant Public Health Physician, UAE

Maryam Al Shamlan, Qatar MoH, Qatar

Shatha Bani Sakhr, Researcher/psychologist, Jordan

Karam Saad Salman, MHPSS Advisor, Iraq

Youth ambassador (YAs)

Purpose: Youth advisors (aged 18-29) with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. YAs bring unique youth-centred perspectives to the development and implementation of project activities.

Members: Alaa AbdelGawad, Climate Change and Health activist, Egypt

Methodology

We produced this research and action agenda through a robust and inclusive methodology to capture, combine and refine a rich diversity of perspectives while fostering connection across a growing community of practice.

The CCM core team developed this methodology in consultation with Regional Communities of Practice (RCoPs), a Global Advisory Board and Wellcome. Methods and materials were adapted regionally to ensure a balance of global standardisation with regional appropriateness and flexibility. Continuous sharing between regions of processes, learnings and challenges facilitated the iterative development of the methodology. The process for developing the regional research and action agendas is shown below (see Figure 1).

Figure 1: Research and action agenda development



Pre-dialogue scoping

Global scoping and framing

- We performed a global-level scoping review of current reviews, key papers and policy reports relating to the climate change and mental health field.^{11,12,13,14,15,16,17,18,19,20,21,22,23}
- We undertook a mapping of the research categories covered by previous reviews and literature.
- We conducted a second mapping of the recommendations for action on climate change and mental health proposed by policy and academic literature.
- We used these results to frame the dialogues and research agenda to align with the current field while responding to key gaps.

Scoping research

- We engaged with lived experience experts through key informant interviews to ensure that the production of the research agenda was appropriately grounded in an understanding of the key lived experience needs emerging in the region.
- We conducted a perspective-gathering exercise to solicit diverse viewpoints from participants in NAWA/MENA who shared their lived experience of climate-related mental health challenges and associated socio-economic and cultural risk factors.
- We undertook a literature review to analyse existing studies, papers and reports on climate change and mental health specifically within NAWA/MENA

Dialogue methodology

- We held two virtual dialogues (3 hours each) with experts across disciplines, sectors and countries in NAWA/MENA.
- Dialogue discussions were designed by the global team, amended to be locally appropriate, and facilitated by members of the RCoP, who are themselves research, policy and lived experience experts.
- Many facilitators actively contributed to the discussions and therefore played a role in both data collection and production.
- The first dialogue identified regional needs and generated research priorities.
- The second dialogue gathered feedback on draft research priorities, identified how to enact research in the region and translate evidence to action in policy and practice, and explored the diversity of regional perspectives and understandings of key relevant concepts.
- The second dialogue involved a spectrum mapping exercise to analyse key concepts like climate resilience, climate anxiety and community engagement. This exercise involved mapping the diverse perspectives of participants on these concepts.
- Data generated in the dialogues included: Google Jamboard notes written by participants, Zoom chats, notes made by dedicated notetakers and transcripts of all discussions.

See Appendix 1 for dialogue agendas.

Survey methodology

We distributed two online surveys to participants and the wider RCoP:

- A **pre-dialogue survey** before Dialogue 1 to inform dialogue design and to solicit perceptions on regional climate impacts, climate-related mental health impacts and research priorities.

- A **post-dialogue survey** after Dialogue 2 to obtain a second round of feedback on research priorities and to identify relevant methods, metrics and datasets to address these priorities.

Full details of the survey methodology are outlined in Appendix 2.

Analysis methodology

- **Research categories:** The Climate Cares Centre conducted a global landscaping exercise of relevant existing climate change and mental health reviews and identified four broad research categories as areas of critical need for further work globally. This framework was used as the basis for structuring discussions within dialogues to generate research priorities and formed the global coding framework for analysis.
- **Analysis:** Dialogue data (transcripts of breakout rooms and notes) was analysed using the Framework Method²⁴ – a matrix-based approach that allows qualitative researchers to undertake deep interrogation of transcripts and written notes. The analyst team populated a matrix based on the global coding framework by creating nuanced summaries with key quotes drawing from the notes and transcripts of the break-out discussions.
- **Draft priority research themes:** This matrix was used to draft a list of priority research themes, refined through consultation with the RCoP and triangulation across breakout notes, transcripts, pre-dialogue scoping and expert consultation. Selection of priority research themes adhered to a globally developed structure and selection criteria (e.g., potential to answer greatest regional emerging needs and evidence gaps, potential to inform decision-making in policy and practice, research feasibility). Further details can be found in Appendix 3.
- **Refinement of priority research themes:** through exercise conducted with Dialogue 2 participants was designed to refine and enrich the research questions within each priority theme. Via a Jamboard exercise, participants were asked to provide feedback on specific research questions by considering the following prompts:
 - What would be most helpful to know about this topic to understand and respond to it?
 - What would you want to know first?
 - Is this a valuable question? Would you change or add anything?
 - Share any ideas on datasets, metrics, or methods that could be applied to help address this.This process was aimed at ensuring the research questions were comprehensive, relevant, and capable of guiding effective responses to the identified themes. Research themes were refined in response to this feedback and shared with dialogue participants and a wider sample of experts in the post-dialogue survey. Participants gave free text responses to the question '*Is there anything you would like to change, remove or add to this research theme?*' and were given the option to suggest additional research themes.
- **Finalisation of priority research themes:** A final list of priority research themes was generated based on incorporation of survey feedback and consultation with the RCoP, regional experts, CCM core team, Global Advisory Board and Wellcome.

Coding frameworks for the research agenda and action agenda can be found in Appendix 4.

Participants

Dialogue participants were a diverse group across geographical spread, gender, sector and discipline. All participants were invited to both dialogues; however, in some cases, participants were unable to attend both dialogues.

In total 50 participants attended Dialogue 1 and 31 participants attended Dialogue 2. The tables below provide a breakdown of participant characteristics.

Table 1: Geographical spread

Country	Dialogue 1		Dialogue 2	
	Number	Percentage	Number	Percentage
Egypt	7	16%	4	17.0%
France	0	0%	1	4.3%
Iraq	8	18.6%	5	22%
Jordan	14	32.6%	5	22%
Lebanon	1	2%	0	0%
Morocco	2	5%	2	9%
Qatar	1	2%	0	0%
Saudi Arabia	2	5%	1	4.3%
State of Palestine	1	2%	0	0%
Tunisia	3	7%	3	13%
United Arab Emirates	2	5%	1	4.3%
United Kingdom of Great Britain and Northern Ireland	2	5%	1	4.3%

Table 2: Expertise

Expertise	Dialogue 1		Dialogue 2	
	Number	Percentage	Number	Percentage
Climate Change	23	33%	13	35%

Mental Health	17	24%	10	27%
Health	22	31.4%	11	30%
Other	8	11.4%	3	8%
I do not know / Prefer not to say	0	0%	0	0%

Table 3 : Discipline

	Dialogue 1		Dialogue 2	
Discipline	Number	Percentage	Number	Percentage
Activism	15	12%	6	7%
Community	15	12%	13	16%
Education	17	13%	8	10%
Expert through my own lived experience	8	6%	5	6%
Funding	3	2%	2	2.4%
Healthcare	16	12%	10	12%
Non-governmental Organisation	15	12%	13	16%
Policy	11	8%	8	10%
Research	27	21%	15	18%
Other	3	2%	2	2.4%

Table 4 : Gender

	Dialogue 1		Dialogue 2	
Expertise	Number	Percentage	Number	Percentage
Men	6	24%	2	12.50%
Women	17	68%	12	75%

Non-Binary	2	8%	2	12.50%
I do not know / Prefer not to say				

Survey participants:ⁱ

Pre-dialogue survey: 68

Post-dialogue survey: 24

Ethics, data collection and storage

This study has been reviewed and given an ethical favourable opinion by the Imperial College Research Ethics Committee (study title: 'Global Dialogues to set an actionable research agenda and build a community of practice in climate change and mental health'; study ID number: 6522690).

Details on data collection and storage can be found in Appendix 5.

Current state and emerging needs for climate change and mental health in Northern Africa and Western Asia

As the climate crisis escalates, more people globally are experiencing related mental health consequences. However, the current evidence base doesn't fully capture these experiences. To develop research themes that ultimately meet the needs of those experiencing and responding to the interconnections between climate change and mental health, it is vital to know: 1) what do people from different backgrounds, contexts and sectors – particularly those with lived experiences of mental health challenges in the context of the climate crisis – report as their experiences, needs and resiliencies, and 2) what evidence do people making decisions and taking actions on the ground need to adequately respond?

This section sets out the context of the research agenda, presenting a synthesis of what we heard through the dialogues, surveys, expert consultations and literature review as key emerging needs for mental health in the context of climate change. We also present regional perceptions and insights from the literature on the current state of evidence to explore which key emerging needs are not yet being captured through research.

ⁱ Please note numbers are approximate and do not account for duplicate or incomplete responses.

Current research on climate change and mental health in Northern Africa and Western Asia

The intersection of climate change and mental health in NAWA/MENA presents a complex challenge, as rising climate impacts and environmental stressors compound pre-existing vulnerabilities for individuals and communities. The Regional Community Convenor could not identify studies from countries in the region that highlight the multifaceted ways in which climate change exacerbates mental health issues in this region, stressing the need for further targeted research and adaptive interventions. Studies have shown that NAWA/MENA is particularly vulnerable to water scarcity and food insecurity exacerbated by climate events such as increased temperatures.^{25,26} These stressors pose a risk to mental health outcomes in NAWA/MENA, underscoring the dire need for comprehensive strategies to research and address these challenges. Furthermore, the exacerbation of political instability due to climate stresses is another risk for NAWA/MENA²⁷. This may also increase mental health burdens, as uncertainty, social unrest, and certainly armed conflict can contribute to psychological distress and increase the prevalence of mental health challenges.²⁸ The indirect effects of climate change, such as economic downturns and forced migrations due to environmental degradation, may also play a critical role in shaping mental health outcomes, highlighting the intersectionality of environmental and social determinants of health in the NAWA/MENA context.

In NAWA/MENA, the intricate interplay between rapid urbanisation and climate change presents challenges to mental health. The migration from rural areas to increasingly congested urban centres, driven by the search for better living conditions and exacerbated by climate-related agricultural disruption, places significant stress on individuals and communities. Urban areas in NAWA/MENA, known for their vibrant but fast-paced nature, can overwhelm newcomers, leading to feelings of isolation and difficulty in adapting. This was further confirmed during expert consultations, with participants highlighting how traditional gatherings – which serve as integral threads of social connection – are increasingly threatened by extreme climatic conditions, leading to a growing sense of cultural disconnection. Climate change compounds these stresses with more frequent and severe weather events, such as sandstorms, floods, wildfires, and extreme heatwaves, directly impacting the wellbeing of both urban and rural populations. Such events not only pose immediate physical dangers but also long-term psychological distress, manifesting as anxiety, depression, and a profound sense of loss and displacement. This scenario was highlighted by the participants during the dialogues who underscored the urgent need for robust support systems to aid in mental health resilience amidst the challenges of urban living and environmental crises, as reaffirmed in the literature.²⁹

Dialogue participants shared that there is currently insufficient research on the specific impacts of climate change on mental health in NAWA/MENA. They emphasized how limited evidence is available to understand the effects of climate change hazards, such as heatwaves, on vital sectors like food and water security, and how these impacts may affect the mental health of the already vulnerable groups.

Framing of key concepts

The overall scope and focus of CCM have been guided by the framing of climate change and mental health outlined in the introduction. 'Climate change', 'mental health' and their intersections and related terms, along with other relevant key concepts, are also understood and defined in diverse ways in NAWA/MENA. Identifying, acknowledging and honouring the ways these terms are understood and

used in different settings is critical to help foster connections, awareness and recognition across disciplines, cultures and communities. This section highlights regionally relevant understandings, as generated through a mapping activity in Dialogue 2, designed to reveal the diversity of perspectives around key concepts, and other discussions throughout the project.

During Dialogue 2, the concept mapping exercise revealed rich, layered interpretations of 'climate resilience,' 'climate anxiety,' and the role of 'community engagement' in addressing climate challenges. The exercise demonstrated the various ways different disciplines and sectors understand these terms.

For example, 'climate resilience' was perceived in multiple ways including:

- Climate resilience was viewed as taking action to build sustainable and healthy communities while focusing on sectors such as infrastructure, health, economy and food systems.
- Other dialogue participants recognized climate resilience as an individual's capacity to cope and manage eco-anxiety enabling them to lead climate action.
- Climate resilience was also seen as shaping policies, technical solutions and financial tools to support the health, mental health and wellbeing of communities during extreme climate events.
- "An ongoing process that requires continuous adaptation" was another definition of climate resilience in the context of health sector adaptation during extreme weather and climate events.

'Climate anxiety' was considered as fatigue and exhaustion from climate events such as extreme heat and drought and their aftermath impacting food availability and augmenting the economic burden of increasing prices. Other participants defined climate anxiety as feelings of hopelessness and fear about the future with the potential of becoming a positive motivator. Similarly 'community engagement' was considered as an essential tool for climate action and mental health support through awareness, education and culturally relevant skill-building initiatives.

Regional needs for mental health in a changing climate

While the climate crisis affects every part of the world, current and future hazards are unevenly distributed and variable. For instance, coastal communities may face increasing rises in sea levels, coastal erosion, and storm surges, while Inuit communities face melting sea ice. Different parts of the same country or region may face different levels of threat to droughts, wildfires, floods, or extreme heat. Understanding the predicted climate hazards and their timescales for different communities and regions is vital to appropriately target support for mental health.

This section outlines the key climate exposures facing NAWA/MENA projected to approximately 2030 as compared to historical baseline (generally 1986-2005), as modelled by climate experts at the Red Cross Red Crescent Climate Centre. We also present findings from the dialogues, surveys and consultations with lived experience experts to explore perceptions of 1) mental health risks associated with these climate exposures, 2) whose mental health may be most at risk, 3) the pathways through which climate exposures might produce or exacerbate existing mental health problems, 4) climate adaptation and mitigation actions that may benefit mental health, and 5) mental health actions/solutions that can help respond to these mental health impacts.

Climate hazardsⁱⁱ

NAWA/MENA is one of the most vulnerable to climate change. The region has regularly experienced severe droughts, wildfires, floods, landslides, storms and extreme temperatures in recent years according to the international disaster database EM-DAT. The region is facing an increase in the frequency and severity of a range of climate hazards, including the following, modelled to approximately 2030 as compared to historical baseline (generally 1986-2005):^{30, 31}

- **Extreme heat** across the region, with Libya and Egypt, in particular, experiencing an increase of up to more than 20 additional hot days per year (high confidence);ⁱⁱⁱ
- **Droughts and wildfires** in the Mediterranean region and northern Africa (low to medium confidence);
- **Heavy precipitation events and flooding**, including extreme precipitation in Egypt, Sudan, Sahel, Sahara and the Arabian Peninsula (high confidence); pluvial flooding over Sahel, Sahara (high confidence) and the Arabian Peninsula (medium confidence); and coastal flooding across the region (high confidence); and
- **Sea level rise** around Africa, contributing to an increase in the frequency and severity of coastal flooding in low-lying areas and to coastal erosion along most sandy coasts (high confidence).

What mental health outcomes appear to be impacted?

Dialogue participants shared a spectrum of mental health challenges arising from climate change, expressed as the mental health symptoms listed below:

- **“Fear and panic during climate events,”** akin to an acute stress disorder, feeling overwhelmed by the immediate threats and long-term implications. This was compounded by “exhaustion from daily change, heat, drought,” akin to chronic stress, impacting not just mental health but also tangible aspects of life like food availability and prices, which in turn may increase stressors on mental health.
- **“Powerlessness,”** with individuals feeling “hopeless and fearful about the future” due to the scale of environmental challenges and a lack of control over climate-related events, with the reported impacts on their lives often reflecting symptoms of depression.
- **Climate anxiety**, resembling generalised anxiety disorder in described symptoms, manifests as a paralysing force and can lead to feelings of futility in daily activities and long-term goals. Notably, concerns about “food security due to extreme weather conditions” and “fear for future income and where to live” painted a picture of the pervasive anxiety that extends to various facets of life, prompting a psychological response that is both a reflection of and reaction to the anticipated impacts of climate change on society and future generations. However, it was also noted that for some, this anxiety could transform into “a positive motivator,” similar to the concept of post-traumatic growth, spurring action and engagement with climate solutions.

ⁱⁱ Information based on Red Cross Red Crescent Climate Centre mapping exercise and projections (methods detailed in Appendix 6). Future projections are based on the middle-of-the-road emissions scenario (SSP2-4.5 Shared Socio-economic Pathway) from the Coupled Model Intercomparison Project - Phase VI (CMIP6) multi-model GCMs ensemble provided in the IPCC, 2021.

ⁱⁱⁱ Confidence levels listed after projections are a measure of researcher confidence that model's results are true and not due to chance.

Who is particularly affected?

Dialogue participants identified the following groups as particularly vulnerable to the mental health impacts of climate change in NAWA/MENA:

- Coastal communities vulnerable to sea level rise, erosion and flooding;
- Displaced populations and migrants;
- Elderly people and children;
- Individuals with pre-existing mental health conditions;
- Rural/agricultural communities; and
- Socially marginalised groups with limited resources or support systems.

What are the pathways and mechanisms linking these climate hazards to mental health outcomes?

Participants identified several critical pathways and mechanisms linking climate hazards to mental health outcomes which include:

- **Impact on livelihoods:** Desertification and rising temperatures undermine the financial stability of workers and farmers, catalysing distress.
- **Food and water insecurity:** Food insecurity and water scarcity can impact mental health, especially in agricultural communities, due to the stress and physical strain associated with water collection and the subsequent threats to nutrition.
- **Displacement:** The movement of populations due to climate change-related events and their aftermath exacerbates mental health challenges such as distress and depression.
- **Social isolation from extreme heat:** The confinement of individuals to their homes and social isolation due to extreme heat significantly increases the prevalence of heat-related stress, anxiety and other mental health challenges.
- **Climate awareness:** Increased consciousness about climate change, including anxiety over future climate events, signifies how anticipatory stress and concern for climate-related disasters lying ahead can directly impact mental health.

What climate adaptation and mitigation actions have benefits for mental health?

Participants identified the following climate adaptation and mitigation actions as beneficial to mental health:

- **Increasing access to green spaces, particularly in low-income urban areas:** Enhancing access to nature and green spaces can provide mental health benefits while also contributing to climate mitigation and adaptation.
- **Enhancing climate resilience through awareness:** Conducting public awareness campaigns that promote the co-benefits of climate actions can empower communities and support their mental health by reducing eco-anxiety and promoting a sense of agency to mitigate climate impacts.
- **Encouraging community involvement:** This involvement can include participation in conservation efforts, community gardens and other sustainable practices.
- **Enhancing infrastructure:** Enhancing infrastructure, such as creating green spaces and outdoor facilities, not only promotes mental wellbeing but also plays a crucial role in mitigating climate-

related health impacts by improving air quality and reducing chronic respiratory diseases. Reducing air pollution is directly linked to lower rates of anxiety, depression, suicide and schizophrenia.

By leveraging these opportunities, it is possible to address both mental health concerns and climate change impacts simultaneously, creating co-benefits for individuals and communities.

What mental health actions can help respond to these mental health impacts?

Participants highlighted several mental health actions to respond to the mental health impacts of climate change, including:

- **Supporting mental health programmes:** Investing in mental health programmes and services, particularly for vulnerable populations, can help individuals cope with the psychological impacts of climate change and build resilience.
- **Conducting research:** Further research on the psychological effects of climate change in specific regions can provide valuable insights for developing targeted interventions and support systems.
- **Policy and community engagement:** By integrating mental health considerations into climate adaptation and resilience measures – strategies designed to enhance a community's ability to withstand and recover from climate-related events – we can improve community wellbeing. This integration involves developing policies and community initiatives that consider mental health impacts when planning for climate adaptation and disaster risk reduction
- **Health system adaptability:** Enhancing the capacity of health systems helps to provide services during extreme climate events.
- **Public awareness campaigns:** Raising public awareness about the link between climate change and mental health fosters understanding and action.
- **Utilising technology:** Exploring the use of telemedicine and digital mental health interventions can enhance access to mental health services and support during climate change-related events.

Further contextual information on regional needs

- **High baseline stress in urban areas:** Urban environments can contribute to heightened baseline stress levels. NAWA/MENA has one of the world's most rapidly expanding populations with millions of people moving from rural areas to cities in search of better economic opportunities, improved living standards, and access to services.²⁵ The physical and social environment in cities, including factors such as heat waves and air pollution, can exacerbate mental health challenges due to the “heat island effect”; this links urban environmental factors to temperature-related mortality associated with mental disorders.³²
- **Vulnerability due to existing water scarcity:** Water scarcity has been linked to mental health outcomes through indirect pathways such as land degradation, famine, and reduced air quality. Lack of water and sanitation can lead to mental health problems stemming from shame, stress due to time lost fetching water and conflicts over water access.³³ Water scarcity can also indirectly impact mental health by exacerbating stressors associated with water insecurity.^{34, 35}
- **Complex social and political dynamics posing implementation constraints:** NAWA/MENA is recognised as one of the world's most water-stressed and vulnerable to the effects of climate change. This vulnerability has potentially negative implications for the region's political stability and security. For instance, the transition from hydrocarbon dependence to renewable energy sources

in oil-rich countries is likely to produce significant political, social and economic ramifications, potentially reconfiguring bi- and multi-lateral relations within the region. Such transitions and uncertainties can also heighten stress, anxiety and other mental health challenges among populations.³⁶

Research agenda

Priority research themes

Background to research categories and priority research themes

This research agenda presents an aligned vision to guide the climate and mental health field in NAWA/MENA. Research priorities have been generated through consultation with experts across disciplines, sectors and geographies in the region and iterated with experts regionally and globally. The priority research themes represent areas where targeted research investment could create a full picture of impacts, mechanisms and solutions across both mental health and climate actions.

Research priorities are presented within four overarching research categories that were identified as critical areas of need for further work globally and that map the climate and mental health research space at a high level, based on an initial review of global literature. Note that some priorities span multiple categories.

- **Impacts, risks and vulnerable groups:** improving our understanding of the extent to which mental health is affected by climate change and for whom. For example: what mental health outcomes are impacted or at risk; the prevalence, severity, economic and societal costs of these impacts; and who is most vulnerable to these impacts.
- **Pathways and mechanisms:** improving our understanding of how climate change affects mental health and, in particular, whether there are factors specific to climate change that increase mental health risks or create new experiences of mental health challenges. This includes considering biopsychosocial or environmental pathways and mechanisms.
- **Mental health benefits of climate action (adaptation and mitigation):** understanding and quantifying when and how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental health.
- **Mental health interventions/solutions in the context of climate change:** identifying the most effective mental health interventions and solutions to support mental health in the context of climate change, across diverse sectors. This encompasses providing support to people already experiencing negative mental health impacts and reducing risk or severity of future negative impacts.

We hope that these priorities will act as an inspiration to guide the direction of the research community of practice, provide investment targets for funders and focus the generation of evidence to best enable policymakers and practitioners to address the emerging and predicted mental health needs in response to climate change in NAWA/MENA.

Priority research themes

Research category	Priority research theme (n = number of research questions)
1. Impacts, risks and vulnerable groups	1. Exploring the nature and prevalence of mental health challenges faced by populations in NAWA/MENA due to the effects of extreme temperatures, including reaching temperatures of 50 degrees Celsius, or heatwaves and evaluating strategies to address these challenges in the short term. 2. Understanding how exposure to heatwaves and extreme temperatures impacts the mental health of vulnerable populations, such as the elderly, children and individuals with pre-existing mental health conditions, and what interventions can address these challenges in affected communities. 3. Identifying the mental health consequences experienced by populations in NAWA/MENA due to the compounding impacts of climate-related hazards (e.g., the combination of increased droughts, electricity shortages and extreme temperatures). 4. Understanding the extent to which climate change (e.g., via more frequent flooding) is likely to increase the prevalence of post-traumatic stress disorder (PTSD) among children and adolescents in inland areas of NAWA/MENA and designing strategies to respond to this increase. 5. Exploring the potential long-term mental health consequences of climate change-related hazards' effects on determinants of mental health – such as loss of livelihoods or increased social inequalities – on marginalised populations in NAWA/MENA countries over the next 10 years. 6. Understanding how climate-induced displacement and forced migration contribute to mental health challenges among affected populations in NAWA/MENA. 7. Understanding how chosen migration driven by climate change-related factors within and between countries in NAWA/MENA impacts mental health challenges and mental wellbeing for individuals and communities and whether there are enduring psychological consequences of this migration. 8. Exploring how climate change impacts the mental health of individuals reliant on agriculture in NAWA/MENA, particularly through changes in crop yields, food security and livelihoods. 9. Understanding the projected impact of climate change on the mental health of coastal communities in NAWA/MENA countries, considering the increased risk of flooding and sea-level rise over the next 10 years.
2. Pathways and mechanism	1. Understanding the distinct cultural understandings of depression (as well as other mental health challenges) in NAWA/MENA compared to Western perspectives, and how these interpretations relate to mental health challenges

in the context of climate change. Guided by these understandings, identifying which culturally sensitive interventions best support mental health in the face of climate hazards.

2. Outlining the mechanisms by which prolonged droughts and water scarcity lead to mental health challenges for individuals and communities in NAWA/MENA and identifying mental health interventions, as well as innovative water management techniques, to mitigate these impacts.

3. Understanding whether loss of identity and cultural disconnection affect the relationship between desertification/land degradation and worsened mental health among Indigenous communities.

4. Exploring the confinement of individuals to their homes due to extreme heat and the resulting social isolation and disruption to outdoor activities (e.g., exercise) as potential pathways by which extreme heat worsens mental health outcomes in NAWA/MENA.

5. Understanding the pathways and mechanisms by which increased awareness about climate change may influence mental health outcomes (for instance, by increasing anxiety, concern and anticipatory stress over future climate events and climate-related disasters). This includes discerning whether particular aspects of 'climate anxiety' are another manifestation of generalised anxiety disorder or whether climate anxiety has aspects that are specific and novel experiences.

6. Mapping other possible biopsychosocial pathways by which climate change affects mental health in NAWA/MENA, exploring their validity and identifying targets in these pathways for intervention.

3. Mental health benefits of climate action (adaptation and mitigation)

1. Understanding the economic, social and mental health co-benefits of investing in renewable energy sources and green infrastructure projects in NAWA/MENA and how these co-benefits can be optimised to improve mental health outcomes, especially in marginalised communities.

2. Exploring how different stakeholders – including policymakers, healthcare providers and local communities – view the integration of climate adaptation and mitigation strategies and mental health initiatives in NAWA/MENA and understanding how to address these perceptions to facilitate effective integration.

3. Understanding the role of traditional knowledge and Indigenous practices in climate mitigation and adaptation interventions that have mental health co-benefits, and how to integrate these practices in the development of culturally sensitive approaches to mental health promotion strategies for diverse communities in NAWA/MENA.

4. Identifying the most effective and evidence-based 'best buy' interventions for addressing both climate change and mental health challenges in NAWA/MENA. Evaluating how policy frameworks and governance structures influence the implementation and scalability of these interventions and determining policy recommendations that promote the integration of these interventions into national agendas.

5. Identifying the most effective interventions to simultaneously mitigate climate change impacts and enhance mental health outcomes across settings, considering factors such as urban/rural divide, socioeconomic status and cultural differences.

4. Mental health interventions/solutions in the context of climate change

1. Developing, implementing and evaluating community-based interventions to address the mental health needs of individuals affected by climate change-related hazards, such as extreme weather events or food insecurity, in NAWA/MENA countries within the next five years.

2. Understanding the extent to which community-based interventions reduce mental health disparities and enhance capacities to adapt to a changing climate among demographic subgroups (e.g., youth, elderly, women) in diverse NAWA/MENA settings, and identifying the underlying mechanisms of their effectiveness.

3. Understanding how culturally tailored interventions build mental health resilience and facilitate climate change adaptation strategies among vulnerable populations in NAWA/MENA countries and identifying the key factors determining the success of these interventions

4. Exploring ways to harness innovative digital technologies and social media platforms to enhance the reach and effectiveness of interventions addressing both climate change and mental health, particularly among youth populations in urban areas of NAWA/MENA.

5. Assessing targeted mental health interventions and climate action initiatives to alleviate the mental health burdens of those facing *displacement* due to climate-related factors.

6. Identifying the gaps in mental health workforce capacity to address climate-related mental health challenges in NAWA/MENA.

7. Identifying the most effective training and capacity-building programs for the mental health workforce to address the mental health impacts of climate change and exploring how to scale and replicate these practices.

8. Understanding how educational institutions in NAWA/MENA can incorporate climate-related mental health challenges in curricula and training programs for mental health professionals and whether graduates are

adequately prepared to address these climate-related mental health challenges.

9. Understanding how to integrate traditional and Indigenous healing practices into training programs for mental health professionals in NAWA/MENA, enhancing their cultural competence and enabling them to effectively support communities dealing with climate-related mental health challenges.

10. Evaluating strategies to recruit, train and retain a mental health workforce – such as psychiatrists, psychologists and social workers – to respond to climate-related mental health impacts on diverse communities in NAWA/MENA.

11. Assessing strategies and policies to enhance the collaboration between relevant disciplines and sectors – such as mental health professionals, environmental scientists, community leaders and policymakers – to effectively address climate-related mental health challenges in NAWA/MENA.

12. Evaluating 1) how digital technologies and telehealth interventions contribute to workforce capacity building to provide mental health services in remote and vulnerable areas affected by climate change in NAWA/MENA, and 2) assessing the challenges and opportunities associated with their development and implementation.

13. Understanding the role of community-based organisations and non-governmental agencies in enhancing the mental health workforce capacity to respond to climate-related mental health challenges in NAWA/MENA and how to effectively coordinate these efforts with governmental initiatives.

Overview of themes for impacts, risks and vulnerable groups

Why were these themes chosen as priorities?

These priority research themes emphasise the perceived mental health impacts of climate hazards like heatwaves, drought and flooding in NAWA/MENA, particularly on vulnerable groups like children, the elderly and marginalised communities. Research to capture, understand and monitor these vulnerabilities is key to developing targeted and effective interventions for climate-related mental health impacts.

NAWA/MENA, already grappling with water scarcity, faces intensified challenges due to climate change. Ensuring water security is not just a matter of physical health and survival but is deeply tied to the psychological wellbeing of communities, making it imperative to consider mental health implications when devising adaptation and management strategies.

Research on the impacts of climate change on agricultural systems, including changes in rainfall patterns, droughts and pests, is essential for developing resilient agricultural practices and ensuring food security. The uncertainty and instability in food production directly contribute to stress, anxiety and uncertainty among farmers and communities, underscoring the need for interventions that address both agricultural resilience and mental health support. By securing food production, we can reduce the mental health challenges posed by food insecurity and economic instability.

NAWA/MENA's long coastline makes it susceptible to the impacts of sea-level rise and coastal erosion. Research into the vulnerability of coastal areas, including risks to human settlements, infrastructure and ecosystems, is crucial for developing adaptation strategies and coastal management plans. Integrating mental health considerations into adaptation strategies is vital to ensure the holistic wellbeing of at-risk populations.

What are examples of methodologies, metrics and datasets that could be used to address these?

There were no identified datasets from the dialogues and surveys that captured the required information for any of the research priority categories. A comprehensive scoping review of relevant datasets in the region could identify relevant data or opportunities to collect such data in the future with key partners. Further investment to routinely collect relevant data is also warranted.

While participants weren't able to identify comprehensive methodologies, metrics, and datasets to address these research priorities, participants suggested the following potential options:

- Methodologies
 - Epidemiological studies to assess the prevalence of mental health conditions.
 - Environmental impact assessments to evaluate climate hazards.
 - Social science research methods for understanding human behaviour and responses.
- Metrics
 - Mental health: Incidence of stress, anxiety, PTSD, depression.
 - Environmental: Frequency and intensity of heatwaves, droughts, floods.
 - Socioeconomic: Rates of migration, changes in livelihood, food security levels.
- Datasets
 - Climate data from meteorological departments for tracking heatwaves and extreme temperatures.
 - Health surveys and medical records for mental health statistics.
 - Agricultural and economic data to assess the impact on livelihoods and food security.

Overview of themes for pathways and mechanisms

Why were these themes chosen as priorities?

Understanding mechanisms linking climate events to mental health outcomes pinpoints strategic points of intervention to disrupt cascading impacts. For instance, if drought affects livelihoods, which then prompts displacement-related trauma, addressing disrupted livelihoods could prevent subsequent mental health harms. Findings from addressing these priority research themes can facilitate tailored, efficient interventions.

Dialogue participants highlighted the need to understand "pathways between these events and mental health" and mechanisms like "loss of livelihood, forced displacement, climate anxiety."

What are examples of methodologies, metrics and datasets that could be used to address these?

While participants weren't able to identify comprehensive methodologies, metrics and datasets to address these research priorities, participants suggested the following potential options:

- Methodologies:

- Comparative cultural studies: To understand the distinct cultural understandings of mental health impacts in general and in relation to climate change compared to Western perspectives.
- Psychological impact assessments and socio-environmental surveys: To understand the mechanisms by which prolonged droughts and water scarcity lead to mental health challenges.
- Impact studies: To assess mental health benefits of renewable energy development.
- Metrics:
 - Mental health diagnostic criteria (which varies by culture).
 - Stress and anxiety levels in relation to environmental factors.
 - Prevalence of mental health disorders in coastal communities.
 - Wellbeing indicators in relation to water scarcity.
- Datasets:
 - Mental health surveys.
 - Environmental data on drought, desertification, and water scarcity.
 - Health records: Tracking mental health trends in affected communities.
 - Climate projections for coastal regions.
 - Water resource and usage data.

Overview of themes for mental health benefits of climate action [adaptation and mitigation]

Why were these themes chosen as priorities?

Understanding mental health co-benefits of climate actions can encourage integrated initiatives and policies across climate, public health and social service spheres. Evidence could inform resource allocation optimize joint outcomes.

The discussion on community engagement in climate action reveals its significant impact on mental health. For instance, one participant emphasized the perceived value of awareness campaigns to potentially empower communities, alleviate eco-anxiety and foster resilience. Another noted, “I see community engagement as more beneficial for mental health than for climate mitigation,” suggesting that active involvement can provide psychological benefits. The input on “providing resources, training and education to community members” underscores the role of skill enhancement in coping with climate-related stress. Furthermore, the idea that “strong community engagement is essential for effective climate action and mental health support” is key to both environmental and mental wellbeing. Balancing these perspectives, these research priority themes centre on assessing the direct mental health impacts of community engagement in climate action.

What are examples of methodologies, metrics and datasets that could be used to address these?

- Methodologies:
 - Stakeholder analysis and attitude surveys: To understand perceptions and barriers in integrating climate and mental health initiatives.
 - Ethnographic studies: For exploring the role of traditional knowledge and practices.
- Metrics:
 - Awareness and attitude levels: Regarding climate mitigation and mental health.
 - Cultural Indicators of Resilience: For traditional knowledge and practices.³⁷

- Climate Change Vulnerability Indices (CCVI): To measure the impact on mental health.³⁸
- Quality of life indicators post-renewable energy implementation.
- Datasets:
 - Survey data: From stakeholders including policymakers, healthcare providers and communities.
 - Climate data and health records: For assessing the impact of climate vulnerabilities.

Overview of themes for mental health interventions/solutions in the context of climate change

Why were these themes chosen as priorities?

Developing appropriate mental health interventions is crucial for equipping health systems to address the escalating climate-related mental health burdens, yet there is a significant gap in our understanding of what interventions are most effective in this context. Preparedness planning and proactive approaches to build resilience and coping skills for handling incremental stressors are assumed beneficial, but empirical evidence is needed to substantiate these approaches and ensure they are tailored to specific regional contexts. The need for interdisciplinary perspectives to achieve a holistic understanding emphasizes the requirement for research to identify and tailor appropriate strategies, rather than taking their effectiveness as a given.

While there is a clear need for interventions providing mental health services, social assistance and practical aid, we need evidence on how best to implement them. The reciprocal links between climate events and mental health suggest a complex interaction that requires further exploration to inform action. Research is needed to validate the necessity of these interventions and to understand the mechanisms through which they can be most effectively deployed, ensuring that action taken is grounded in evidence of what works in specific contexts and for different vulnerable populations.

What are examples of methodologies, metrics and datasets that could be used to address these?

- Methodologies:
 - Comparative intervention studies: To identify mental health co-benefits of effective climate adaptation and mitigation actions.
 - Cultural adaptation research: For understanding the impact of culturally tailored interventions.
 - Community-based participatory research: To assess community interventions' effects on mental health disparities.
 - Digital and telehealth intervention analysis: Evaluating digital platforms for mental health service delivery.
- Metrics:
 - Mental health outcomes: Measuring changes in mental health status post-intervention.
 - Community engagement levels: Assessing participation in, and response to, interventions.
 - Digital engagement metrics: Usage and effectiveness of digital technologies for mental health support.
- Datasets:
 - Mental health service data: Records of mental health interventions and outcomes.
 - Climate data: Information on climate hazards and mitigation efforts.

- Community demographics and health records: Data on diverse groups within NAWA/MENA.
- Digital platform usage data: Statistics on the use of digital technologies for mental health.

Action agenda

As the climate change and mental health field in NAWA/MENA builds and the evidence base grows, it is crucial to avoid perpetuating existing challenges, including 1) disconnections across disciplines and between researchers and policymakers; 2) unequal focus on topics and geographies; and 3) and siloed decision making for climate and mental health. This action agenda sets out a shared vision as a rallying focus of the mental health and climate change field in NAWA/MENA. This includes guidance on how to best support the growing community of practice, how to translate evidence to action, and the principles that should guide this approach. Enacting this agenda will require transdisciplinary effort and coordinated action across research, research funding, policy, and practice. This action agenda aims to guide this work by setting out the challenges that must be addressed, opportunities that can be harnessed, and priority actions to work towards a thriving climate and mental health field.

Regional vision for mental health in a changing climate

Vision statement

Our vision is a NAWA/MENA region where the interconnections between climate change and mental health are widely understood and incorporated into policy and practice. Communities facing climate threats here will have access to resources and support systems protecting their mental health and wellbeing and fostering resilience.

Key characteristics of the desired future state for climate change and mental health in the region

The desired future state includes mainstream recognition of climate-mental health links in health systems and public discourse. Vulnerable communities have access to targeted mental health services during climate events and actively build psychological resilience skills. Mental health training equips practitioners to address climate-related burdens. Co-beneficial initiatives promote environmental sustainability and mental wellness synergistically. Climate adaptation and health policies are developed collaboratively to maximize mutual gains. Public attitudes openly acknowledge mental health challenges prompted by eco-grief and anxiety, reducing stigma.

Creating an enabling environment for research at the intersection of climate change and mental health

The identified research priorities will only be of value if they are enacted. The climate and mental health field is relatively new and rapidly growing, and now is the time to ensure that it is designed to deliver a mentally healthier future in the context of the climate crisis. In a field that spans multiple disciplines and sectors, each with different cultures and ways of working, and on a topic with low awareness in many countries, what is needed to support capacity-building efforts? What principles must guide the field, and

what are the challenges and opportunities in the region to create an environment that would enable such research?

This section presents a synthesis of dialogue discussions and survey results on what is needed to implement the research agenda and foster an enabling research environment for climate change and mental health.

A desired state of climate change and mental health research in NAWA/MENA

The desired future state includes mainstream recognition of climate-mental health links in health systems and public discourse. Vulnerable communities have access to targeted mental health services during climate events and actively build psychological resilience skills. Mental health training equips practitioners to address climate-related burdens. Co-beneficial initiatives promote environmental sustainability and mental wellness synergistically. Climate adaptation and health policies are developed collaboratively to maximize mutual gains. Public attitudes openly acknowledge mental health challenges prompted by eco-grief and anxiety, reducing stigma

Challenges holding back research

Lack of region-specific metrics

Participants emphasized the importance of developing region-specific metrics and measures that are compatible with the needs of NAWA/MENA. While specific details were not provided, the participants emphasized the need to move away from blindly following metrics and measures derived from the United States and Anglo-Saxon literature.

To develop region-specific metrics and measures, it is crucial to consider the unique characteristics, challenges, and priorities of NAWA/MENA. This may involve conducting research and analysis that considers the region's specific climate conditions, socio-economic factors, cultural contexts and political dynamics.

While not raised in the dialogues themselves, some potential areas where region-specific metrics and measures could be developed include:

- **Emissions and adaptation metrics:** Develop measures to assess the psychological effects of high emissions and climate vulnerability, such as scales for climate-related stress and anxiety. Additionally, metrics for evaluating the mental health benefits of adaptation measures, including indices of psychological relief and community wellbeing improvements, are needed.
- **Health impact measures:** Create indicators for the prevalence and severity of mental health issues directly related to climate change impacts, utilising culturally relevant depression and anxiety scales. This includes measures to evaluate the effectiveness of mental health interventions and policies tailored to climate impacts.
- **Infrastructure and resilience indicators:** Identify metrics that quantify the mental health contributions of resilient infrastructure, such as the reduction in stress and anxiety levels afforded by stable access to water and energy during extreme weather events. This could involve developing scales for measuring perceived security and wellbeing linked to infrastructure reliability.
- **Socioeconomic factors metrics:** Establish measures to examine the impact of socioeconomic factors on mental health in the climate change context. This could include metrics for assessing

coping mechanisms and mental resilience against climate-induced stressors, as well as the mental health improvements from enhancing socioeconomic resilience.

- **Public awareness and engagement measures:** Design metrics to link public awareness and engagement in climate action to mental health outcomes. This may involve creating scales to measure the psychological impacts of feeling informed and involved in climate action, including the reduction of eco-anxiety and the enhancement of community cohesion.

It is important to note that developing region-specific metrics and measures requires collaboration and input from local stakeholders, including researchers, policymakers, communities and civil society organizations. This participatory approach ensures that the metrics and measures are contextually relevant, culturally sensitive, and aligned with the specific needs and priorities of NAWA/MENA.

Additional challenges

Dialogue participants noted several other challenges equally holding back research in NAWA/MENA, including:

- Limited resources and funding aligned with regions' vulnerabilities;
- Stigma around mental health;
- Lack of awareness among professionals and the public of the issue of climate change and mental health;
- Lack of updated datasets for both climate change and mental health and the relation between them;
- Political issues and intricacies influencing research and interfering with implementation;
- Cultural and ideological resistance to change; and
- Lack of harmonising perspectives between collaborators.

Opportunities and enablers

Multiple opportunities and enablers were identified in the dialogues to enable the climate and mental health research field in NAWA/MENA, as outlined below:

- **Building awareness:** Enhancing awareness of the climate change and mental health nexus and where research is needed for the climate-mental health field at both global and regional levels.
- **Sustaining the CCM Regional Community of Practice:** Facilitating dialogue between specialist knowledge and community realities to inform research on climate change and mental health.
- **Collaborating with universities to integrate climate and mental health in curricula:** Integrating climate change and mental health education would foster the development of students into researchers, stimulating their interest and involvement in the climate-mental health field and enhancing their multidisciplinary research capacity across both areas.

Relevant potential partners

Climate change and mental health issues cut across traditional areas of expertise and sectors of society, making integrated partnerships among a wide range of stakeholders an imperative for research in this field. Recommended partners are included in the list below.

- **A diverse array of research partners spanning multiple disciplines, including:**

- **Climate science:** Provide data and projections on climate change impacts, helping to establish the link between climate change hazards and mental health outcomes.
- **Epidemiology:** Conduct studies to determine the prevalence and incidence of mental health disorders related to climate change and identify the factors that contribute to their development.
- **Environmental science:** Study the physical and ecological changes caused by climate change and their potential effects on mental health.
- **Policy and governance:** Analyse the existing policies and frameworks related to climate change and mental health, identify gaps and propose evidence-based policy recommendations.
- **Psychology and psychiatry:** Provide expertise in understanding the psychological impacts of climate change on individuals and communities, as well as diagnosing and treating mental health disorders.
- **Public health:** Assess the population-level impacts of climate change on mental health, identify risk factors and develop strategies for prevention and intervention.
- **Social sciences:** Provide insights into the social and cultural factors that influence mental health in the context of climate change, including social inequalities, community resilience and adaptation strategies.
- **International organisations**, such as UNICEF and the World Health Organisation.
- **Local stakeholders and organisations**
 - Adolescents and youth.
 - Local community leaders.
 - Schools and universities.
 - Women's groups.
- **National stakeholders and organisations**, including government ministries and agencies.

Priority next steps/recommendations to investors and actors

Enacting this research agenda will require transdisciplinary approaches with collaboration across stakeholders. The priority next steps for investors and actors in the region include:

1. **Establishing and sustaining a regional climate and mental health network:** Utilise CCM as a foundational platform to bring together experts, stakeholders and communities. This network would facilitate the exchange of knowledge and capacity-building efforts, streamline access to funding opportunities for the identified research and action priorities, share best practices and develop regional strategies tailored to the unique challenges and opportunities within NAWA/MENA.
2. **Securing commitment from key stakeholders:** Engage national actors, international organisations and government ministries to commit to a unified strategy that prioritises mental health in the context of climate change. This involves formalising partnerships and securing funding for research and intervention programs.
3. **Developing research and policy interventions and projects that address the climate-mental health nexus:** Invest in capacity-building efforts to empower individuals and organisations with the knowledge and skills to design interventions that address climate and mental health. Initiate pilot projects that exemplify the integrated approach to climate change and mental health. These

projects should focus on critical areas, such as enhancing community resilience, developing culturally sensitive mental health interventions and implementing sustainable environmental policies.

By taking these steps, investors and actors can significantly contribute to building a robust response to the mental health challenges posed by climate change in NAWA/MENA, ensuring that efforts are well-coordinated, evidence-based and culturally sensitive.

Translating a growing evidence base into action that can respond to the mental health impacts of climate change

The current evidence base on the interconnections between climate change and mental health compels action in policy and practice to protect escalating mental health needs and promote co-beneficial climate actions. How can current evidence and new insights created through implementing this agenda best translate into changes to policy decisions and practices across both climate and mental health regional spaces?

This section presents a synthesis of dialogue discussions and survey results, setting out the challenges and opportunities to translate evidence generated through research into policy and practice.

A desired state of climate change and mental health to action in NAWA/MENA

We envision a rapid translation of findings through collaborative partnerships spanning research, policy, health practice and community sectors. Evidence-based interventions and policies evolve continually as the knowledge base grows. Consideration of climate-mental health interlinkages in regional and national policy frameworks is mainstreamed. Together these create an enabling environment for evidence-based decision-making and implementation of strategies that promote resilience.

Challenges holding back translation of evidence

The diverse challenges identified by participants in the MENA dialogues include:

- Lack of coordination across sectors;
- Limited funding for implementation-focused research;
- Climate change scepticism, which hinders political will;
- Mental health stigma, which obstructs open dialogue and help-seeking behaviours;
- Lack of public awareness of the interlinkages between climate change and mental health to draw policymakers' attention and prompt action;
- Overburdened health systems with little capacity to integrate additional considerations; and
- Burnout among practitioners working with distressed communities.

Opportunities and enablers

Dialogue participants identified multiple opportunities and enablers for action on climate change and mental health in NAWA/MENA, as outlined below:

- Strong civil society networks provide infrastructure for grassroots action and social support;
- Youth increasingly vocalising climate-mental health concerns, building momentum for change;
- Expanding access to technology and digital mental health solutions, offering scalable interventions;
- Regional commitments like Iraq's Sanadi framework, which demonstrate receptiveness to progress; and
- Ongoing disasters, which while tragic, provide impetus for transformation while rebuilding.

Relevant potential partners

Identified potential partners by participants of the MENA RCoP include:

- **Academic institutions:** Research universities like the American University of Beirut and King Saud University.
- **Corporations:** Major companies in the region, such as Aramco in Saudi Arabia, for funding and resources.
- **Government ministries:** Ministries of health, environment, and education in countries like Egypt, Saudi Arabia and Jordan.
- **Health systems:** National health services and major hospitals in the region for intervention delivery.
- **Intergovernmental agencies:** Regional WHO offices and UN branches in NAWA/MENA for technical guidance and support.
- **Media organisations:** Media companies that broadcast in multiple countries in several languages such as MBC, TRT and BBC.
- **NGOs:** The Red Crescent Societies and various local NGOs for community engagement.

Collaboration across this ecosystem is vital to address the complex societal challenges at the climate-mental health nexus.

Priority next steps/recommendations to investors and actors

To translate the evidence base on climate change and mental health into policy and practice, the following call to action is proposed.

Policymakers should:

- Recognise and integrate the interlinkages between mental health and climate change identified by the CCM regional research and action agenda within national health and environmental policies.
- Regularly assess climate vulnerabilities including climate-related mental health impacts.

Funding bodies should:

- Allocate resources for implementation research and metrics development on climate change and mental health.
- Invest in workforce education and training for scaling up interventions that respond to the mental health impacts of climate change.

Research and academic institutions should:

- Conduct research to identify service provision gaps.
- Foster platforms for cross-sectoral partnerships and knowledge exchange.

Community stakeholders and organisations should:

- Develop campaigns to address climate skepticism and mental health stigma and raise awareness of the significant interlinkages between climate change and mental health in the region.
- Collaborate with schools and universities to integrate climate change and mental health education into curricula.
- Provide climate and mental health awareness and capacity-building to parents and families.

Discussion: strengths, limitations and next steps for the research and action agenda

Strengths

Strengths of this research and action agenda include its inclusion of diverse disciplinary expertise and lived experience. This foundation allowed collaborative co-creation of a regional vision and priorities for progress. The proposal of implementation activities in addition to research priorities makes this agenda uniquely actionable. Its alignment with existing regional commitments provides a pragmatic starting point to integrate related activities with other strategies, such as the IOM's Middle East and North Africa Regional Strategy 2020-2024 and global initiatives like the [United Nations Framework Convention on Climate Change \(UNFCCC\)](#) and its Paris Agreement.

Limitations

Limitations include a lack of specificity regarding funding opportunities for suggested next steps. Additionally, while vulnerability insights were drawn from the literature and dialogue participants, primary research would provide a vital nuanced understanding of distinct regional needs to inform localised solutions. The dialogues involved representatives from across sectors, including those involved in policy spaces, but did not include policymakers directly. Including this group could help to secure imperative political buy-in.

Next steps

Key next steps for advancing this research and action agenda involve its widespread distribution among essential stakeholders, including national government bodies, funding organisations, health leaders across both practice and research domains and climate scientists. We aim to solicit feedback, nurture a sense of collective ownership, and galvanise action-oriented commitments toward the articulated priorities. The establishment of regional communities of practice through the convening of these varied groups can create a solid foundation for continuous dialogue, the building of strategic relationships and

the alignment of efforts across sectors.

Drawing on the success of the CCM project in Jordan, we recommend leveraging the project's experiences and outcomes to inform and accelerate similar efforts regionally. The project's engagement with the Ministry of Health in Jordan – which led to the inclusion of mental health in the Ministry of Health's national strategy and the initiation of a national dialogue (including policymakers and health workers from the private and public sector) on the impact of climate change on mental health – serves as a compelling model. Furthermore, the establishment of pilot clinics dedicated to individuals mentally affected by climate change showcases a practical approach to translating research into actionable health interventions. These accomplishments underscore the potential for CCM to contribute significantly to the field, offering a framework for engaging national stakeholders, informing policy development and pioneering service provision models that can be adapted and replicated in other contexts.

Conclusion

This research and action agenda represents a critical initial step in advancing approaches to address escalating climate-related mental health challenges across NAWA/MENA. It provides a foundation and starting point for strategic collaboration among stakeholders working towards the realisation of a holistic vision of mental health and climate resilience. However, realising demonstrable progress requires dedicated leadership, resources and political will alongside sustained, inclusive participation from all sectors of society. This document seeks to inspire by demonstrating the possibilities if we collectively leverage knowledge, technology, policy, and community strengths. The path forward entails surmounting complex obstacles and system transformations requiring persistent effort. By taking ambitious first steps grounded in compassionate optimism, despite limitations, we can move towards a future where the health of both people and the planet are valued as interconnected.

Hearing from the Regional Community of Practice

- "First of all, there should be awareness about the importance of this topic so that people understand what it is and can participate in it... It should be simple awareness so that people can understand the importance of this topic."
- "We can also raise awareness about mental health through schools. By having counsellors in schools who can educate children and future generations that mental health is a natural part of overall health."
- "It's possible through social media pages, short advertisements on social media, or even on the radio... having an official page about this topic and through posts, discussions, meetings, mental health and other forums we can reach a larger audience."
- "One possible challenge we might face is that not everyone is interested in mental health, meaning the challenge could be that people not understanding this topic is the first challenge... Lack of interest and understanding are the two possible challenges we may face."
- "In rural areas especially, you can find that people may feel ashamed to talk about mental health issues openly... shyness about this issue is a very sensitive matter and many people are hesitant to discuss it."
- "Depression is everywhere... People are thinking about migration... very high temperature reaching to 50 and more."
- "Mild to severe depression, stress and burnout syndromes and anxiety... the three most symptoms regarding mental illness that I have seen."
- "Elderly, children and pregnant women of course are the most vulnerable groups."
- "The most impacted by the hazard would be people with a short of resources... Also perhaps those who work outdoors."
- "It was really affecting a lot of people due to this shortage of water... we have to rationalise the use of water."
- "The climate itself we are speaking about another level. Another level of strategy. It's national strategy... including the aspect of health and mental health in the strategy..."

Glossary of terms

For a glossary describing relevant concepts and key words for the Connecting Climate Minds research and action agendas, please download from [here](#).

Who produced this report

Author contribution statement

The authorship team for this project includes a diverse group of professionals, each contributing their unique expertise to the development and completion of the work. Dr. Yaroup Ajlouni served as the project manager and regional conveyor, orchestrating the team and liaising with professionals across NAWA/MENA. Dr. Duha Omari took on the role of main coordinator, overseeing the distribution of tasks, compiling the work, and leading the content creation and research agenda drafting. Dr. Ashraf Al-Qudah was responsible for reviewing and overseeing the research analysis, in addition to proofreading and reviewing the final agenda. Dr. Nabaz Al Mirani and his team from DAMA Organization acted as regional co-conveyors, enhancing the dialogues, contributing to the regional scoping, and context understanding. Dr. Tariq Al Jarrah facilitated the dialogues and assisted in recording lived experiences. Mo'nes Qudah and Dr. Tala Dabbas played key roles in communication, drafting the agenda, and facilitating discussions, while Baraa Haikal was instrumental in facilitating dialogues and translating content. This team's collaborative effort was pivotal in shaping the project's direction and outcomes. Adding to this team, Alaa Abdelgawad, our Youth Ambassador, was crucial in engaging the youth community and amplifying broader voices, significantly contributing to the facilitation, note-taking, and dissemination of project contents across the region.

This is the independent work of the authors with the support and input of the Connecting Climate Minds team.

Acknowledgements

The content and intellectual property of this report have been produced through a collaborative process with experts across Northern Africa and Western Asia. The authors are so grateful for the diverse expertise, voices, and ideas of those who have contributed to the content presented in this research and action agenda. Accountability for the content of this report lies with the authors, but we share ownership of this work with all these contributors. We would like to acknowledge the contributions made by:

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Funding

Funding for this work was provided by Wellcome.

Conflicts of interest

The authors have no conflict of interest to declare.

Appendix

Appendix 1: Dialogue agendas

Dialogue 1

Time (mins)	What	How
0-20	Welcome and introductions	
20-35	Visioning Exercise	Visioning Exercise: "Headlines" on Climate Centre Good Games platform Participants share what they are hoping to get from being part of Connecting Climate Minds, and why they are invested in this topic <i>Imagine we are in 2030 and the CCM project has received an important prize for its achievements in bringing CC/MH together in your region and making change for climate and mental health together through research and action. You are interviewed by a journalist as you are celebrating winning the prize.</i> Create / vote on and review Headlines created on the Good Games Platform
35-50	Scene setting	Brief context to the links between climate change and mental health What are we hearing in the region that we can come together and respond to: sharing highlights of pre-scoping Presentation on Connecting Climate Minds and Q&A
50-95	Breakout session 1: Regional needs	Host sets scene for breakout discussion Share climate hazards overview for the region Breakout groups Facilitator to lead the group in a discussion on the mental health challenges arising from climate change in the region. Questions posed to participants were: • From your perspective, what are the climate hazards that concern you the most in terms of mental health and wellbeing? • For each hazard, discuss the following: ○ Who is being impacted? Are there any groups more at risk?

		○ What are the mental health and wellbeing outcomes associated with this hazard? ○ How does this hazard specifically affect mental health and wellbeing; what are the mechanism e.g. housing, livelihoods etc ○ Encourage people to look at different types of mechanisms e.g. social, biological, cultural, environmental ○ Climate adaptation/mitigation actions - what is being done, what can be done? ○ Actions that are being/could be taken to improve or promote mental health and wellbeing in this context Brief plenary highlights
95-105	Break	
105-110	Sneak preview	The cartoon artist shares one first draft reflecting some of the content of the session so far. Host invites participants to share in the chat how the image resonates.
110-160	Breakout session 2: Evidence gaps and research priorities	Breakout groups [same participants, facilitator and note-taker as first breakout session] Looking back at the hazard table from Breakout session 1 as a starting point, participants are guided to start thinking about where more research is needed. Hosts invite 2 or 3 facilitators to share some brief reflections
160-170	Cartoon reflection on dialogue output	Participants individually review each cartoon, and share their reflections. This can range from: suggestions for changes to make it more accurate, to sharing how it relates to our reality, or any other insights.
170-180	Wrap-up and next steps	Host shares a high level recap of the session, thanks everyone and celebrates this first coming together of the regional community. Participants complete brief survey to provide feedback on the dialogue

Dialogue 2

Time (mins)	What	How
0-5	World Map Exercise	Host asks participants to place a “star” icon where they currently are and then take the “heart” icon and place it where they would like to visit one day or a place they have been and really loved/feel especially connected to.
5-15	Welcome and introductions	
15-30	Sharing updates	Presentation on global and regional progress and outcomes of dialogue 1, including vision, priority research themes.
30-50	Priority Research Theme Feedback	Jamboard shared on screen. Host introduces priority research theme activity and asks people to consider the following questions: Research questions ● <i>Is this a valuable theme? Would you add or change anything?</i> ● <i>Share any ideas on datasets, metrics or methods that could be applied to help address this.</i> Research topics ● <i>What would be most helpful to know about this topic to understand and respond to it?</i> ● <i>What would you want to know first?</i> Music plays and participants work independently on Jamboard at their own speed.
50-55	Scene setting for breakouts	Host sets the scene for the breakout discussion. Interested in the 'how', which includes the principles and desired outcomes for research, policy and practice in the region. The discussion will enable targeting of action and investment to both implement the research agenda in the region, and to ensure evidence on the climate and mental health nexus is translated into policy and practice

55- 90	Break out 1	Eight breakout groups. The first 4 on the theme “Creating knowledge through research” (e.g. How can we best implement the research agenda?) and the other 4 on “Fostering evidence-based policy and action” (e.g. How can we translate the growing evidence base to action in policy and practice?). Each group uses a Jamboard to discuss: 1) What does this look like when it is done well? 2) What opportunities & enablers exist? 3) What challenges must we overcome? 4) Who should our partners and stakeholders be? Brief plenary highlights
90-100	Break	
100-105	Draw your feelings	Host welcomes everyone back and invites all who'd like to join to grab a piece of paper and a pen/marker (or digital) - and draw how they are feeling today. Anyone willing is invited to show their image - share some reflections.
105-125	Spectrum Mapping	Host introduces and explains the activity. Share screen of Jamboard. Designed to reveal the diversity of perspectives and options around different concepts and to organize them into a meaningful spectrum. This activity aims to illuminate the group's range of perspectives and to organize those perspectives into a continuum. Question: <i>What do these concepts mean to you?</i> Focusing on one board at a time, host asks everyone to silently generate a point-of-view around that concept and write it on a sticky note on the Jamboard. People can add more than one. Once the sticky notes are posted, the host works with the group to sort them into a horizontal range of ideas. Sticky notes that express similar perspectives or options should go next to each other. Sticky notes that seem to be outliers should stand alone; they may sometimes end up defining the limits of the range. Repeat with the remaining concepts. At the end, ask for observations and insights on the lay of the land. And if any perspective or option has been excluded. If so, add it and re-sort as necessary.

125 - 155	Break out 2	Host introduces the second round of breakout rooms and provides a 2-minute summary of the Jamboard contributions from the first breakout. Original 8 groups reconvene and expand and add to their initial thoughts. Groups are allowed to continue on the theme they were already discussing or switch to the other theme.
155- 175	I will ... I want...	Host asks everyone to personally reflect about how they will contribute to this space, and share an "I will" statement in the chat, but only press enter when told to do so. After a minute, host invites everyone to press enter at the same time to make statements appear. This is repeated with a statement about "Out of this community I would like to see..."
175	Wrap up and next steps	Host shares a high level recap of the session, thanks everyone and outlines next steps. Participants complete brief survey to provide feedback on the dialogue

Appendix 2: Survey methodology

The pre-dialogue survey was intended to gather demographic information about the participants, their awareness of, and personal or professional experience with climate change and/or mental health issues. Accessibility requirements were also asked to ensure that dialogues could be designed to suit the needs of participants. The post-dialogue survey was intended to (a) collect additional feedback on the draft research topics after they had been reviewed based on discussions held during Dialogue 2, and (b) collect feedback on the project as a whole. Both surveys were implemented on Qualtrics, and the link was distributed via email.

The pre-dialogue survey was sent to dialogue participants only, while the post-dialogue survey (sent out after Dialogue 2) was sent to both dialogue attendees and the wider regional community, composed of individuals who had expressed interest in receiving project updates. Survey recipients were also asked to forward the post-dialogue survey to others within relevant networks. For those who had not attended the dialogue(s), the invitation to complete the survey included a Participant Information Sheet and a digital consent form. The dialogue participants had already consented as part of the dialogue process.

Appendix 3: Criteria for selection of priority research themes

Priority research theme selection criteria was generated based on discussions with experts globally across sectors, and disciplines, and by drawing on selection criteria used in other research priority-setting exercises.³⁹ The criteria was applied by consultation within the analysis team.

- Does the research theme answer an existing evidence gap?
- Does the research theme have the potential to inform decision-making in policy and practice?
- Does it reflect one or more of the following:
 - identified the greatest mental health challenges linked to climate change in the region.
 - identified the greatest climate hazards in the region.
 - evidence policymakers most need to address emerging needs / overcome barriers to action.
 - alignment with what the experts by lived experience voiced as challenges in the region.
- Would it be feasible to undertake this research?
- Is it framed to be understandable, specific, and practical for the research community to meaningfully address? Could a researcher or a funder with relevant expertise understand this theme?
- Is it specific enough to be meaningful and guide focused research (if too broad or vague it won't help target resources meaningfully)
- It helps funders invest in this area (consider what is regionally relevant here)

Appendix 4: Coding frameworks for the research agenda and action agenda

1. Coding framework for research agenda

Research Category	Sub-categories
1. Impacts, Risks and Vulnerable Groups This category is about improving our understanding of the ways in which mental health is affected by climate change. For example, what mental health outcomes are impacted or at risk, the prevalence, severity, economic and societal cost of these impacts, and who is more vulnerable to these	<u>Cross cutting considerations to keep in mind for all sub-categories:</u> Timeframe Geographical variation
	1.1. Research that focuses on the prevalence, severity and nature of the experience of different mental health outcomes/challenges/experiences affected by different aspects of climate change. This may include research to understand the emergence of climate-specific mental health experiences and their relationship to already defined mental health challenges.
	1.2. Quantifying the fraction of mental health burden (including mortality) caused by climate change.
	1.3. Understanding the risk factors to mental health that are caused or affected by climate change as well as protective factors .

impacts. This category also includes the ways we can go about getting this improved understanding of mental health impacts of climate change - the methods and metrics we need to assess and monitor mental health in ways that are relevant to climate change, contextually appropriate, comparable etc.	1.4. Identifying population sub-groups (e.g. demographics, livelihoods, life stage, pre-existing mental health challenges) who experience increased vulnerability to mental health challenges caused by climate change, and conversely those experiencing resilience to these effects (i.e. vulnerable groups). 1.5. Quantifying the cost (e.g. economic, social) of the additional mental health burden caused by climate change and insufficient climate action. 1.6. Methods research to identify the most appropriate ways to assess and monitor the mental health impacts of climate change [including adapting pre-existing scales, creating new ones, determining appropriate mental health metrics and indicators for inclusion in global processes like Lancet Countdown]. This can also include the need for cross-cultural validation and development of culturally appropriate metrics.
2. Pathways and mechanisms This category is about improving our understanding of <i>how</i> mental health is affected by climate change. We are interested in research themes that can help identify, categorise and understand the range of ways that climate change or climate action may act to affect mental health. This can include considering pathways and mechanisms that are biological, psychological, societal or environmental in nature, and may be direct or indirect. Note that mechanisms can include mechanisms to the development, maintenance, and/or	<u><i>Cross cutting considerations to keep in mind for all sub-categories:</i></u> <i>What factors are linked with increased vulnerability or increased resilience for the associated mental health outcomes.</i> 2.1. Categorising and understanding the societal mechanisms by which climate change negatively impacts mental health [e.g. changes to livelihoods, disruption to cultural practices, food and water insecurity, forced migration, political factors] 2.2. Categorising and understanding the environmental mechanisms by which climate change negatively impacts mental health [e.g. air pollution, reduced exposure to biodiversity] 2.3. Categorising and understanding the psychological mechanisms by which climate change negatively impacts mental health [e.g. how temperature affects cognitive changes relevant to symptoms of mental health challenges]. 2.4. Categorising and understanding the biological mechanisms by which climate change negatively impacts mental health [e.g. impacts of psychotropic medication on thermoregulation, neurodevelopmental factors]. 2.5. Understanding mechanisms whereby climate action or mental health interventions benefit climate and mental health (i.e. co-beneficial mechanism).

resolution of mental challenges, so this includes also mechanisms relevant to guide development or understand workings of interventions	2.6. Methods research to identify the most appropriate ways to assess and monitor pathways and mechanisms by which climate change negatively impacts mental health and wellbeing (e.g. systems mapping across disciplines)
3. Mental health benefits of climate action [adaptation and mitigation] This category is about how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental health. This includes quantifying costs and benefits of climate action for mental health, understanding what is needed to support better alignment between climate action and mental health action, and identifying where this integration is already happening across strategies and policies.	3.1. Identifying climate actions that integrate or align with mental health benefits [co-beneficial climate actions, e.g. increased tree cover in cities]
	3.2. Quantifying co-benefits of climate action for mental health (including number of people experiencing the benefit, size of effect, economic considerations).
	3.3. Exploring how the mental health costs and benefits of climate action may differ across population sub-groups (e.g. demographics, livelihoods, life stage)
	3.4. Understanding the governance structures/decision support tools that enables alignment of action for climate change and for mental health across sectors
	3.5. Mapping and monitoring the integration of mental health within adaptation and mitigation policies across sectors [e.g. National Adaptation Plans, energy, transport, food, water, agriculture]
	3.6. Exploring opportunities for mental health to be integrated into other climate priority areas i.e. loss and damage and climate finance.
	3.7. Determining best approaches for climate action (e.g. emissions reductions or climate adaptation) within the mental health sector (ensuring psychiatric facilities can be kept cool in heat waves; green space projects in mental healthcare facilities)
	3.8. Methods research to identify the most appropriate ways to assess and monitor mental health benefits of climate action [e.g. place-based approaches, methods for attributing and quantifying co-benefits, methods for assessment of the mental health implications of decisions in other sectors]
4. Mental health interventions/solutions in the context of climate change This category is about identifying the most effective mental health interventions/solutions to support mental health in the context of climate change. This might be about	<u>Cross cutting considerations to keep in mind for all sub-categories:</u> LEVEL (e.g.) Individual, Family, Community, Systems MECHANISM (e.g.) Biological, Psychological, Social, Environmental SECTOR (e.g.) Education, Healthcare, Public Health Effectiveness considerations include impacts across different population groups, and implementation considerations might include providers, cost and time.
	4.1. Understanding different ways of knowing, being and doing in different cultures and communities that can build individual, community and ecological resilience

providing support to people already experiencing negative mental health impacts, or about reducing risk or severity of future negative mental health impacts. This includes learning from knowledge held by different disciplines, communities and cultures, understanding how existing mental health interventions are affected by climate change, identifying and evaluating existing interventions that are relevant to the context of climate change, and developing new interventions. Interventions are relevant at all levels (individual, family, community, systems) and across sectors.	4.2. Understanding how existing mental health interventions are affected by climate change
	4.3. Identifying and evaluating mental health interventions that are already designed for or relevant to the context of climate change and/or integrate climate change considerations
	4.4 Amending, implementing and evaluating relevant mental health interventions from other settings to be appropriate for climate-related impacts?
	4.5. Co-designing, implementing and evaluating novel interventions that meet climate-related mental health needs
	4.6. Comparing cost-effectiveness, implementation considerations, and effectiveness across interventions for a particular setting and particular population group to determine "best buys"
	4.7. Identifying, developing and evaluating approaches to awareness-raising and capacity building to upskill workforces to recognise and respond to the mental health impacts of the climate crisis (e.g. mental health professionals, emergency responders)

2. Coding framework for action agenda

Action Category	Sub-category
1. Creating an enabling environment for research at the intersection of climate change and mental health	1.1 Desired state of research - This code captures <i>what good looks like</i> for climate change and mental health research in the region that implements the research agenda. - What are the features of the kind of research that is desired or valuable? Are there specific attributes or milestones that would signify this state of research?
	1.2 Opportunities and enablers

	This code captures opportunities to progress the climate and mental health research field in the region towards the desired state, and factors that would enable progress. May be general or specific, and may link to what is required to overcome the challenges outlined in the next code.
	1.3 Challenges holding back research This code captures challenges that are stopping the climate and mental health field in the region from currently being in the desired state, or are predicted to emerge in trying to create investment in and implementation of the research agenda.
	1.4 Partners/stakeholders This code captures any key individuals, organisations or stakeholder types identified as being particularly important to engage for implementation of the research agenda in the region and securing required investment.
	1.5 Priority next steps/recommendations to investors and actors - This code captures the concrete next steps that need to be taken as priorities to create the conditions in the region to implement the research agenda. - This section will be used in the agenda to inform potential investors and key actors/decision makers where the priorities should be for next steps.
2. Translating a growing evidence base into action that can respond to the mental health impacts of climate change	2.1 Desired state of evidence to action in policy and practice - This code captures <i>what good looks like</i> for action on climate change and mental health in the region based on current and future evidence. - What are the features of the kind of pathways for translating evidence into action that are desired or valuable? Are there specific attributes or milestones that would signify that evidence-based action is occurring?
	2.2 Opportunities and enablers This code captures opportunities to progress evidence-based action on climate and mental health in the region towards the desired state, and factors that would enable progress. May be general or specific, and may link to what is required to overcome the challenges outlined in the next code.
	2.3 Challenges holding back action This code captures challenges that are stopping desired actions to protect mental health from the climate crisis in the region, or to enable co-beneficial climate action. The code may also include

	challenges that are predicted to emerge in trying to ensure that current and future evidence translates into change on the ground and at all levels of policy and practice.
	2.4 Partners/stakeholders This code captures any key individuals, organisations or stakeholder types identified as being particularly important to engage for translation of evidence into relevant action and securing required investment.
	2.5 Priority next steps/recommendations to investors and actors - This code captures the concrete next steps that need to be taken as priorities to translate the emerging evidence base on climate and mental health into action in policy and practice. - This section will be used in the agenda to inform potential investors and key actors/decision makers where the priorities should be for next steps.

Appendix 5: Data collection and storage

Dialogues

Dialogues were conducted virtually on Zoom. Dialogues and breakout groups were recorded and transcribed by third party providers (Way with Words and Absolute Translations) and Zoom chat comments were saved. In Dialogue 1, Word documents were used to capture notes from breakout discussions. In Dialogue 2, Google Jamboard was used to capture notes and for participants to directly contribute comments.

Surveys

Survey distribution and data collection was carried out using the online platform Qualtrics. All survey data was collected by Imperial College London and anonymous data shared with Jordan Health Aid Society International for analysis.

Data storage and sharing

Data was stored and managed by Imperial College London using a secure server. Jordan Health Aid Society International was a Joint Data Controller for the data provided to this project for MENA and responsible for securely storing and sharing data with Imperial College London and with regional analyst teams. Data will be stored by Imperial College London for 10 years after study completion.

Appendix 6: Climate hazards in NAWA/MENA^{30, 31, 40}

Observed climate-related hazards in NAWA/MENA and projected changes

To guide informed discussions on the current and potential mental health consequences of current and future climate change in the region, it was vital to ensure a grounding in climate science. The Red Cross Red Crescent Climate Centre (RCCC) conducted a mapping of previous and projected climate hazards across the region to inform the dialogues and research agenda. The following is a summary of the climate-related hazards that NAWA/MENA have experienced over the last 30 years, and their projected increases to approximately 2030.

It is important to emphasise that the data gathered for the past and future periods differ not only in spatial resolution but are also based on different underlying principles. The former are through observations or documented occurrences such as those reported by national weather services and are aggregated at a country-scale. The latter are from climate modelling-based studies and assessment reports published by the Intergovernmental Panel on Climate Change (IPCC), which are at a more granular spatial scale. For the same reason, the categories of climate-related hazards in the past and future periods may at times not be identical, but are generally comparable or related. For instance, flooding that is reported as a relevant climate hazard in the last 30 years here is not a direct output of global climate models (GCMs). Instead, the underlying meteorological variable (rainfall) from the simulations of GCMs are used for deriving indices that can be considered as a proxy for potential flooding in a future warming climate.

NAWA/MENA

NAWA/MENA is one of the most vulnerable to climate change. The region regularly experiences severe droughts, wildfires, floods, landslides, storm and extreme temperatures in recent years according to the international disaster database EM-DAT (<https://www.emdat.be/>). Note that the limited number of extreme events recorded (e.g. droughts, wildfires, storms and extreme temperatures) may be due to the missing records in the EM-DAT data for various reasons, such as poor or limited network of satellite and surface observations over the region, or limited reporting from national meteorological services. In addition, the spatial aggregation of the data to country-scales makes it difficult to map the occurrence of the past hazards to human settlements. The total number of people affected as reported in EM-DAT therefore may not always be truly representative of the frequency of occurrences of the underlying climate hazard. For instance, a country in NAWA/MENA may have experienced frequent heatwaves, but if these events were not recorded or in uninhabited regions, the total number of occurrences discussed below may be an underreporting.

Previous Droughts

Sudan and Turkey experienced the highest occurrences of droughts and wildfires from 1991-2022, with such six such events occurring in Sudan affecting 4.4 million people on average. Similarly, Turkey has recorded five such events in the same period affecting 140,500 people on average (Figure 1, top). The population affected^{iv} by drought in particular was highest in Iraq, with 7 million people affected on average within this period.

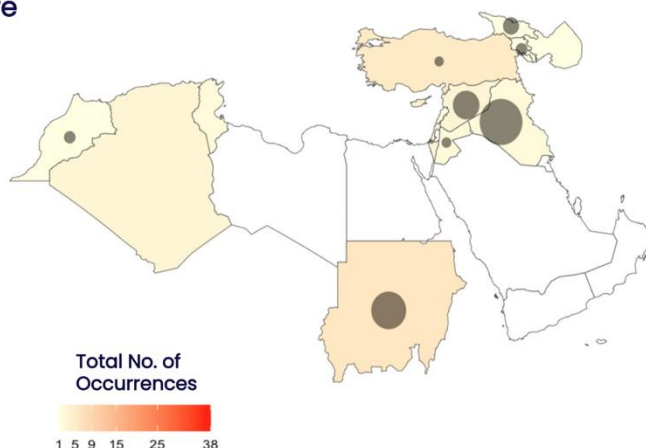
^{iv} Note, EM-DAT definition of “total affected” accounts for “the total of injured, affected, and homeless people”.

Drought and Wildfire

Total Occurrences & Avg.
Affected Population:
1991 – 2022
(Source: EM-DAT)

Average No. of people
affected

- < 10K
- 100K
- 250K
- 1M
- >5M

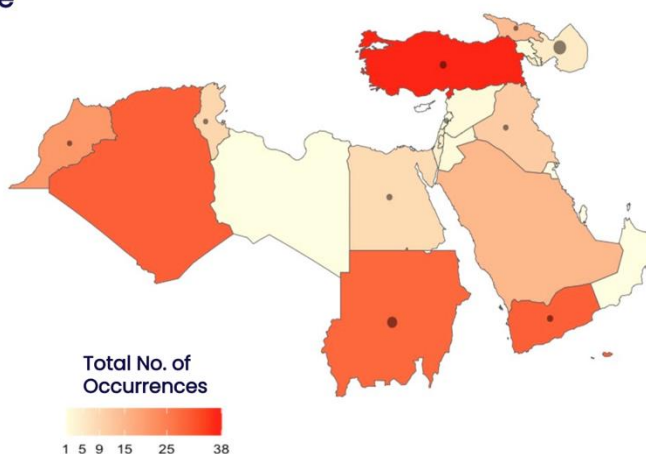


Flood and Landslide

Total Occurrences & Avg.
Affected Population:
1991 – 2022
(Source: EM-DAT)

Average No. of people
affected

- < 10K
- 100K
- 250K
- 1M
- >5M



Storm and Extreme Temperature

Total Occurrences & Avg.
Affected Population:
1991 – 2022
(Source: EM-DAT)

Average No. of people
affected

- < 10K
- 100K
- 250K
- 1M
- >5M

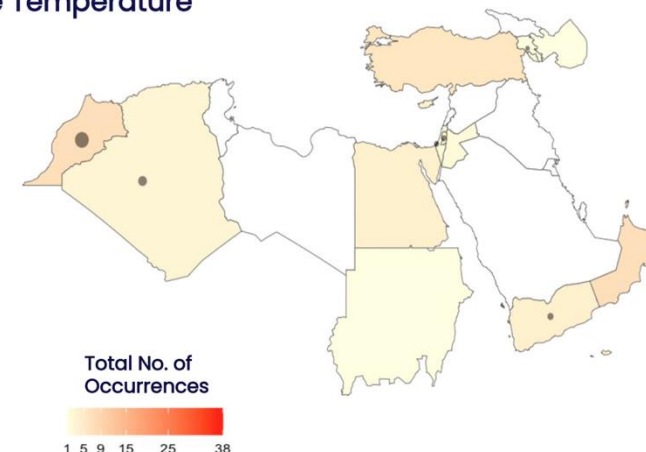


Figure 1. Number of reported droughts and wildfires (top), flood and landslides (middle) and storms and extreme temperatures (bottom) between 1991-2022. Red legend depicts the total number of occurrences of each category of hazard between 1991-2022, the black circle size is representative of the average affected population in that period.

Previous Floods

The number of floods and landslides experienced was particularly high in Turkey where 38 such events occurred between 1991-2022 affecting about 49,000 people (Figure 1, middle) on average during this

period. Furthermore, the floods and landslides in Algeria and Yemen were also particularly high where 30 events were recorded within the same period in these nations, affecting a combined 62,000 people on average.

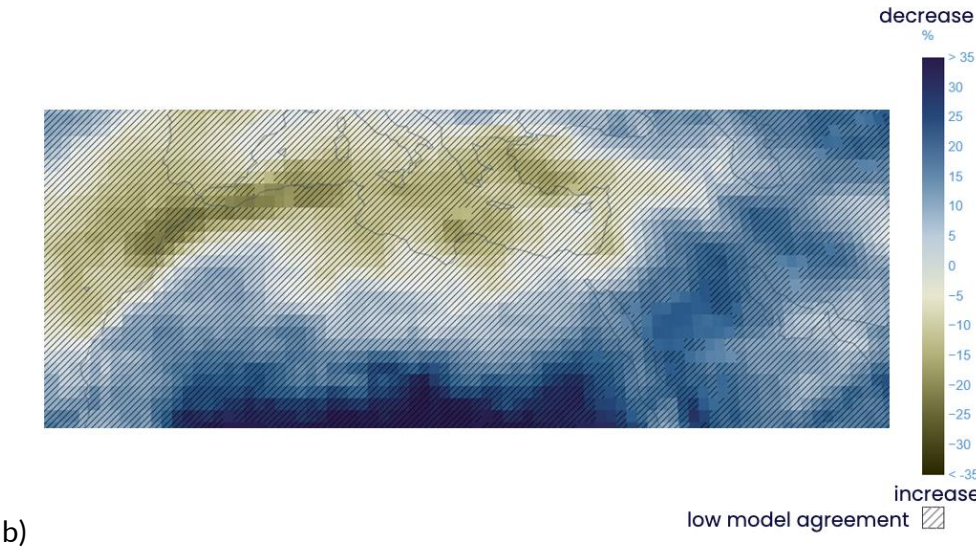
Previous Extreme (Hot) Temperatures and Storms

Morocco and Oman were countries most affected by storms and extreme temperatures within this region with seven such meteorological events recorded to have occurred between 1991-2022 (Figure 1, bottom) in both countries. Morocco's population was also affected most in the region by such events, with an average of 495,000 people reported to have been affected during this period.

Future Projections

Future projections are based on the middle of the road emissions scenario (SSP2-4.5 Shared Socio-economic Pathway) from the [Coupled Model Intercomparison Project - Phase VI](#) (CMIP6) multi-model GCMs ensemble provided in the IPCC, 2021. While not the worst-case emissions scenario, SSP2-4.5 assumes that the 2015 Paris agreement commitments are not achieved and 2°C global warming is not avoided.

- Mean and extreme temperature: The future average change in the number of hot days above 35°C projected for 2030 compared to 1986-2005 show Libya and Egypt will experience an increase of up to more than 20 additional hot days, in addition to the rest of the region experiencing a projected increase of hot days (high confidence) (Figure 2a).
- Droughts and wildfires: Projected changes in the average occurrence of droughts in 2030 show an increase in these natural hazards in the Mediterranean region and northern Africa as well as an increase in fire weather and aridity relative to 1986-2005 (low to medium confidence) (Figure 2b).
- Heavy precipitation events and flooding: Projections show a robust increase in extreme precipitation for higher warming levels (later years) in Egypt, Sudan, Arabian Peninsula (high confidence) (Figure 2c). These projections show an increase in heavy precipitation and pluvial flood over Sahel and Sahara (high confidence), as well as a projected increase in pluvial flooding in Arabian Peninsula (medium confidence).
- Sea level rise: Relative sea level rise is likely to continue around Africa, contributing to increase in the frequency and severity of coastal flooding in low-lying areas and to coastal erosion and along most sandy coasts (high confidence).



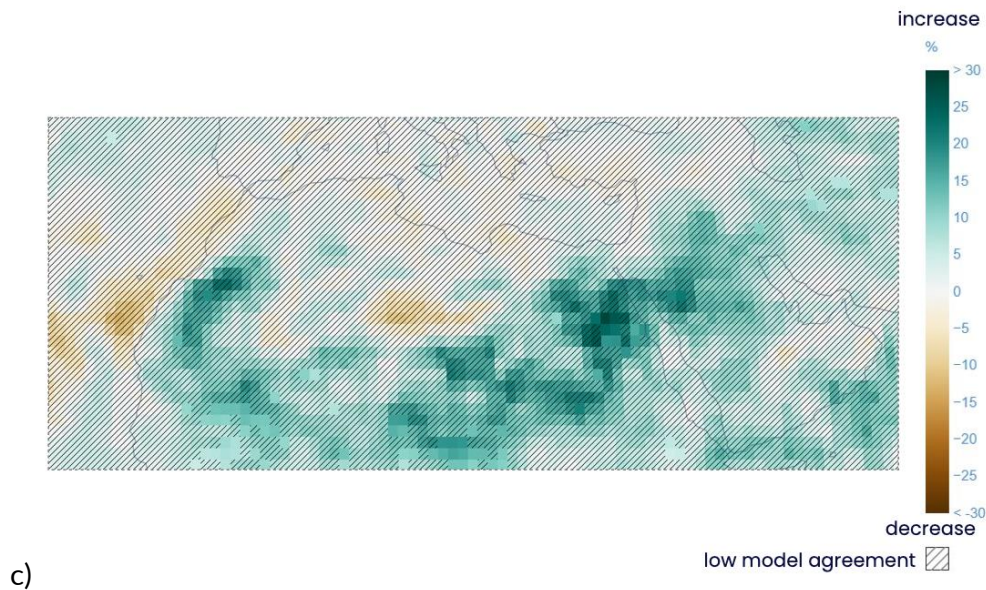


Figure 2. Projected changes in a) hot days above 35°C b) drought index SPI-6 and c) annual maximum 5-day precipitation for 2030 compared to 1986-2005 under the middle of the road emissions scenario SSP2-4.5.

Note: Detailed information [on the projections and the associated confidence](#) were accessed from the IPCC Working Group I contribution to the Sixth Assessment Report "Climate Change 2021: The Physical Science Basis" and the associated [IPCC WGI Interactive Atlas](#) and regional factsheets: [Fact Sheets | Climate Change 2021: The Physical Science Basis \(ipcc.ch\)](#)

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